# Common Denominator

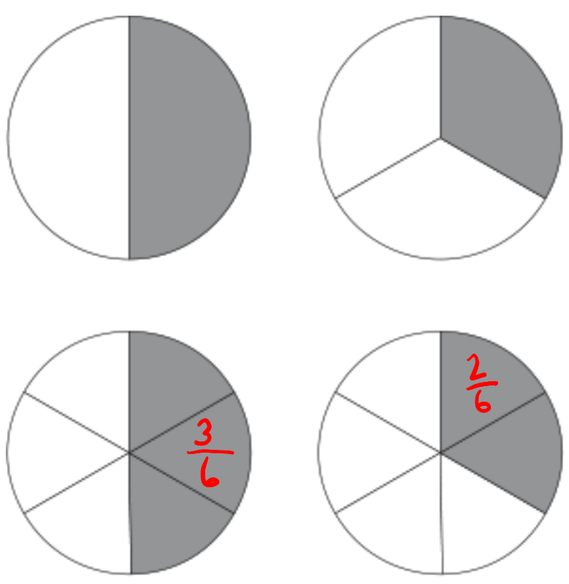
You can add and subtract fractions only if they have the same denominators.

Imagine trying to add $\frac{1}{2}$ and $\frac{1}{3}$. The circles are cut into sections of different sizes, so there isn't a way to combine them.

To add them, look for a way to cut both circles into sections of the same size.

Here the circles are cut into sixths. $\frac{1}{2}$ becomes $\frac{3}{6}$. $\frac{1}{3}$ becomes $\frac{2}{6}$.

Now you can add: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$




In this lesson, you will practice finding a common denominator and writing equivalent fractions.

## Example 1

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You can always find a common denominator by multiplying the denominators.

Find a common denominator for $\frac{3}{4}$ and $\frac{1}{6}$, and write equivalent fractions.

**Step 1**

Multiply the denominators.

$4 \times 6 = 24$

$\frac{3}{4} = \frac{3\times6}{4\times6} = \frac{18}{24}$

$\frac{1}{6} = \frac{1\times4}{6\times4} = \frac{4}{24}$

**Step 2**

Write equivalent fractions with the denominator 24.

$\frac{3}{4} \times \frac{6}{6} = \frac{18}{24}$

$\frac{1}{6} \times \frac{4}{4} = \frac{4}{24}$

Multiplying the denominators isn't the best way to find a common denominator, but it will always work.

The best way to find a common denominator is to find the **least common multiple (LCM)** for both denominators.

## Example 2

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List the first few multiples for each denominator. Then look for the lowest number that both lists have in common.

Find a common denominator for $\frac{3}{4}$ and $\frac{1}{6}$, and write equivalent fractions.

**Step 1**

List a few multiples for both numbers.

**4:** 4, 8, 12, 16, 20, 24, . . .

**6:** 6, 12, 18, 24, . . .

**Step 2**

Find the least common multiple.

The lowest number in both lists is 12.

this is the lowest common denominator (LCD) for this case.

**Step 3**

Write equivalent fractions with the denominator 12.

$\frac{3}{4} \times \frac{3}{3} = \frac{9}{12}$

$\frac{1}{6} \times \frac{2}{2} = \frac{2}{12}$

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